



User Manual

Kunak Device Configurator

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Revision history

Date	Revision	Changes
03/03/2017	1.00	Initial Release
14/05/2018	1.01	Modbus Sensors Update
07/09/2023	2.00	Security Update

Table 1 – Change log

1. INTRODUCTION

The Kunak Configurator application is a wizard style application that guides the user through the necessary steps in order to have the Kunak device ready to communicate data.

1.1 Systems requirements

- Operating system: Windows Vista, 7, 8, 10. There are packages for both 32 and 64 bits versions
- CPU: any capable of running the aforementioned operating systems mentioned above.
- RAM memory: 50 MB
- Hard disk space: 100 MB
- USB port: needed to connect the Kunak device

1.2 Supported Kunak devices

The following Kunak devices are configured using the Kunak Configurator application:

- K1xx: Kunak device with GPRS radio interface
- K2xx: Kunak device with LORA radio interface
- K3xx: Kunak device with Wi-Fi radio interface
- KM2M: Kunak device with GPRS radio interface and serial interface for AT commands

2. CONFIGURATION

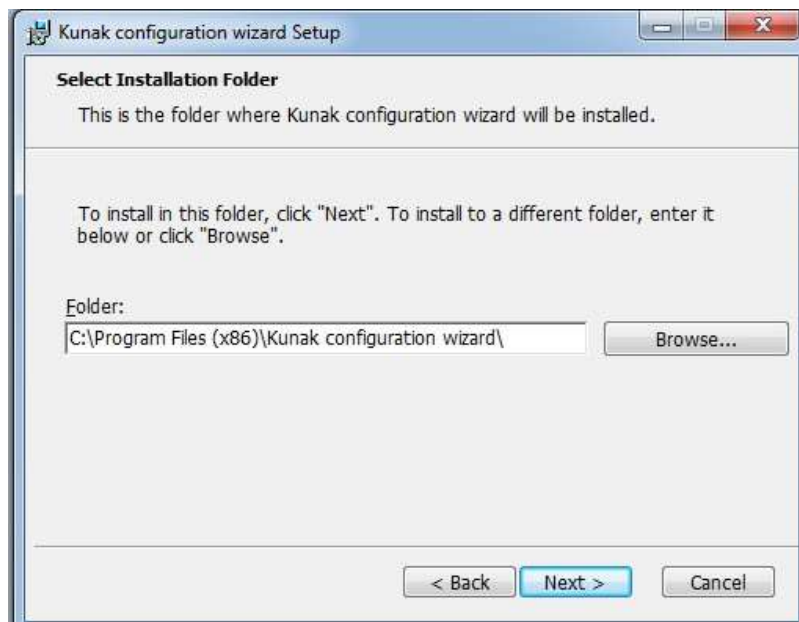
2.1 Kunak Configurator setup

The setup procedure is a very simple wizard style application that guides the user through the setup steps. This section will explain each of the setup wizard pages.

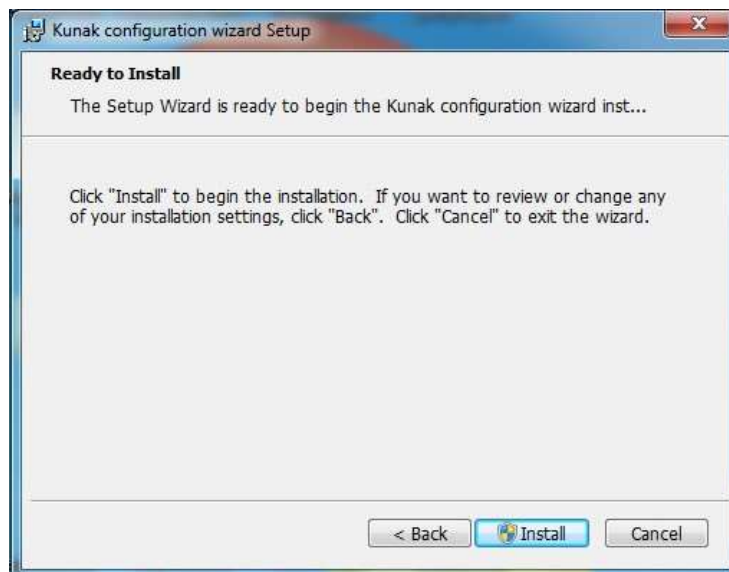
The first step is a window of welcome to the installation:



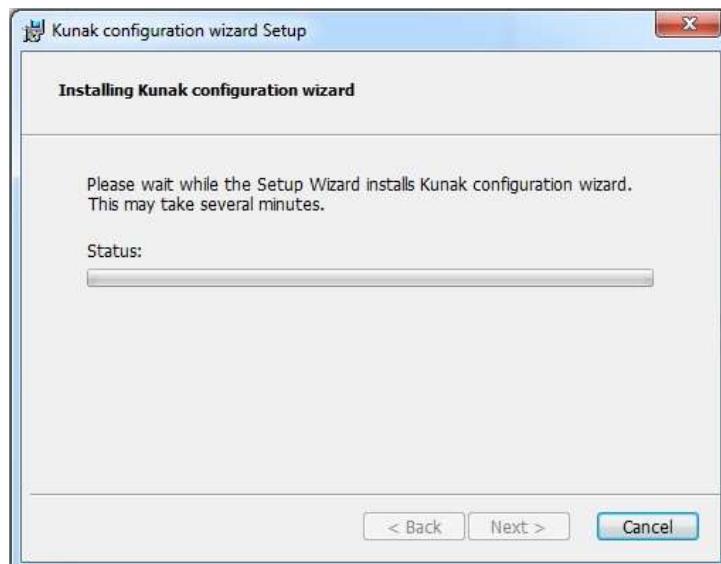
The second step prompts the user for the destination folder.



In the third step, when the 'Install' button is pressed, the installation starts:



Once the 'Install' button is pressed, the setup wizard installs the Kunak Configurator in the desired folder.



While Kunak Configurator is installed a window appears to install the drivers. The Accept button must be clicked to install them.



Finally, the following window will appear on screen confirming a successful installation.

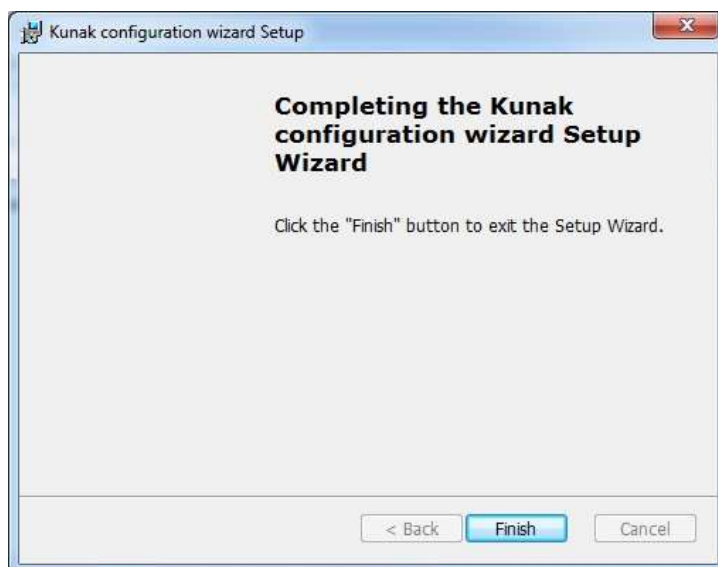


Figure 1 – Successful installation

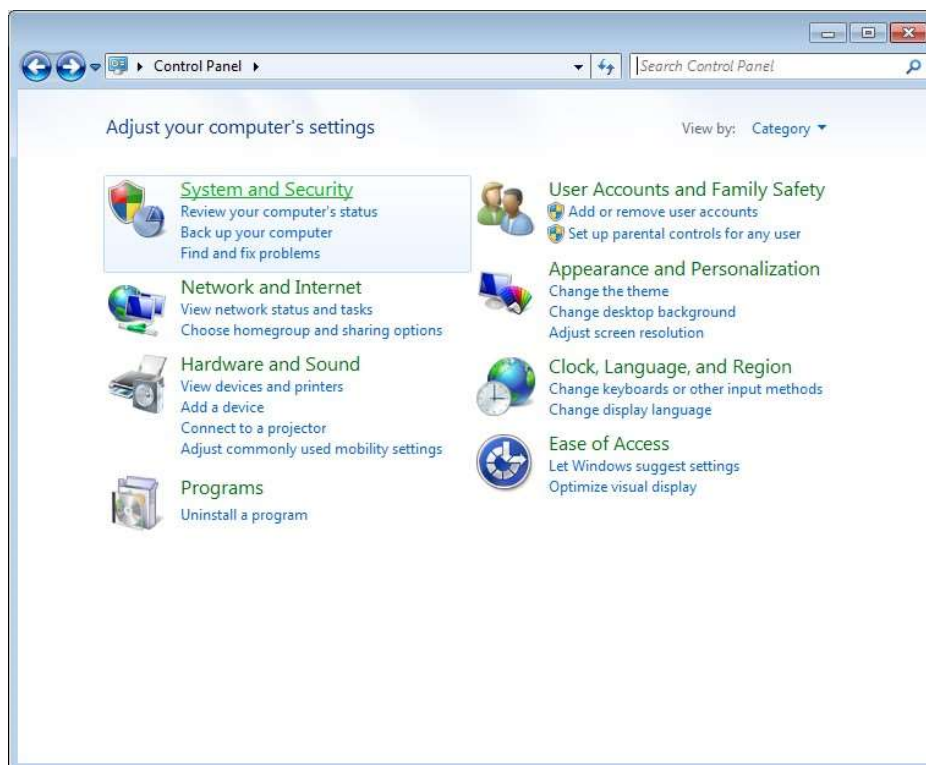
2.2 Manual drivers setup

In case the drivers have not been installed correctly, the following steps must be followed to install them.

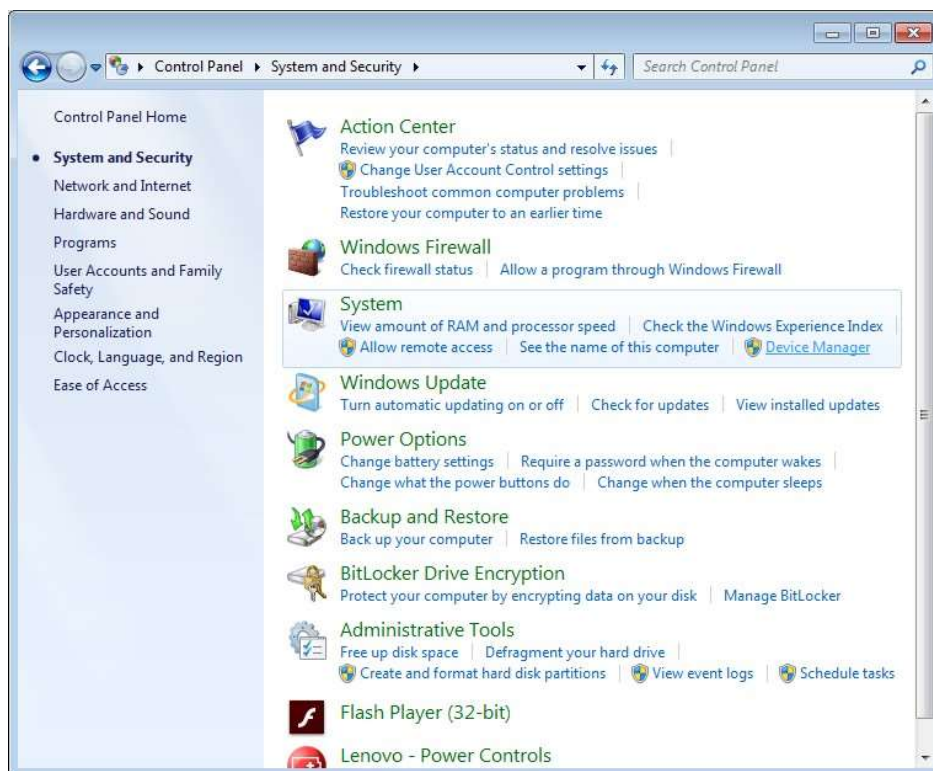
Inside the destination folder, the wizard creates a drivers folder where all the necessary drivers for Kunak products can be found.

In order to install the device driver, follow the steps below:

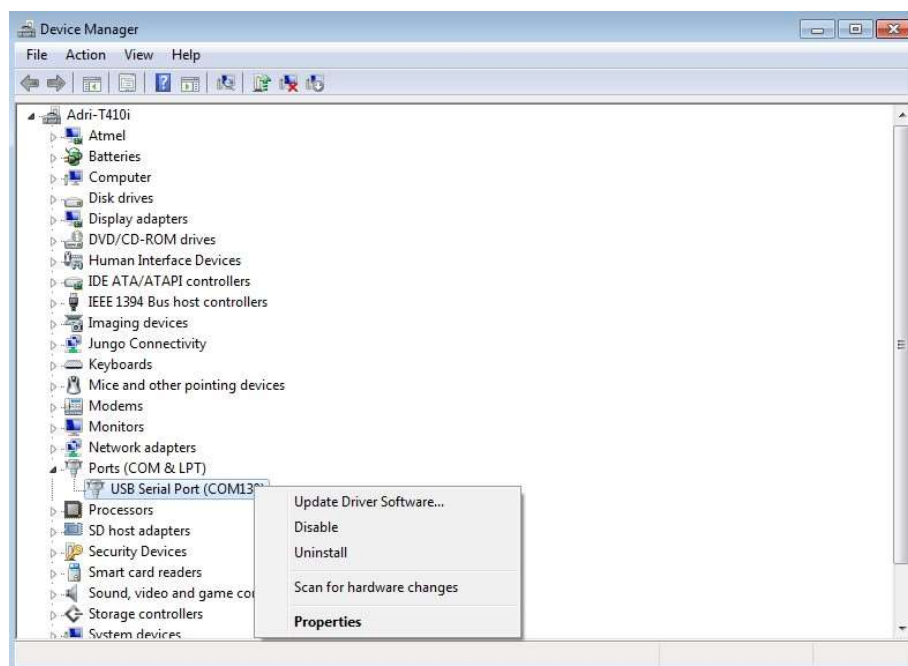
1. Go to Start > Control panel: Once in control panel, left mouse click on System and Security



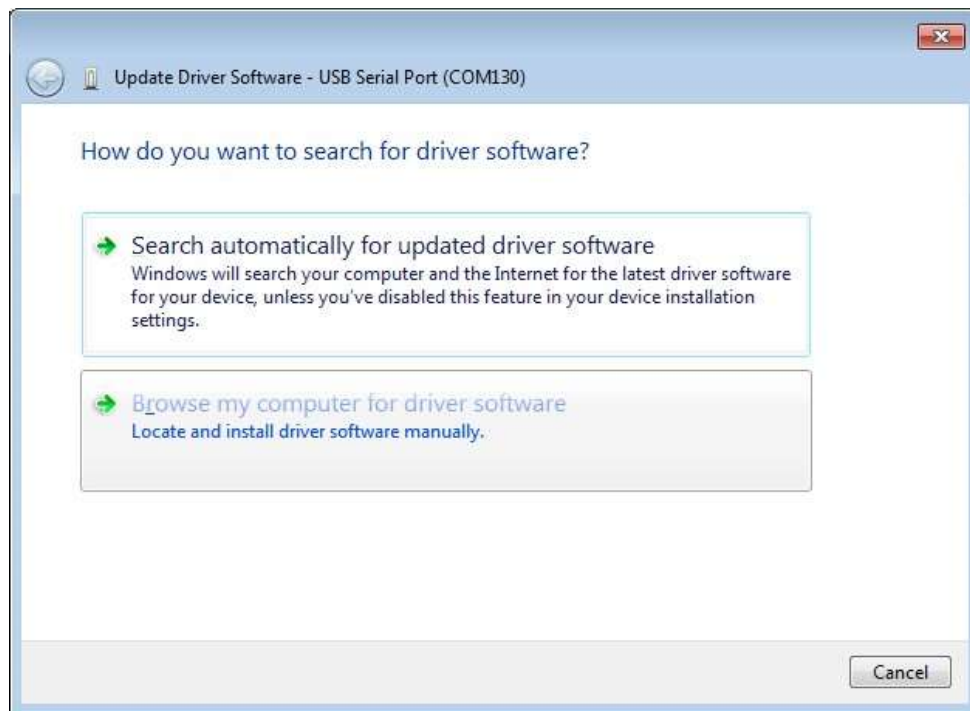
2. Under System and Security, click on Device Manager



3. Once the device manager window opens, plug the USB cable into the device and power it up. The device will show up in the list, under the Ports (COM & LPT) section with a yellow icon the first time it is plugged. Right click on the icon and click the option Update Driver Software in the contextual menu.



On the Update Driver Software window choose the option Browse my computer for driver software.



A text box with a path to the location where the driver is stored will appear. Please, choose the drivers folder located in the Kunak Configuration installation folder (Usually in the Program Files folder) and click Next.



4. The wizard now finds automatically a suitable driver for the device. If the process fails to complete, it might be because the device entered the normal operation mode before the user was able to install the driver. Our devices wait in configuration mode for about 30 to 60 seconds before entering normal mode and turning the USB peripheral off to reduce power consumption. If this was the case, please, reboot the device and start the process again.

2.3 Kunak Configurator

2.3.1 Initial Setup

To configure a device using the Kunak Configurator application, the device must be in configuration mode. Check the specific manual of the device to get details on how the configuration mode works. For detailed information, the manual for the device under configuration can be found on the web section 'Support' (top right), accessible through the following URL: https://kunakcloud.com/user_area/support.jsp.

To run the Kunak Configurator application, just click on the shortcuts either located at Start Menu > All programs > Kunak Device Configurator or desktop. The first step in the wizard is an introductory page that presents the application to the user. It can be skipped right away.

2.3.2 Initial Setup

The 'Detecting devices' screen, shown in the Figure 1, searches among the computer serial ports for a connected device and prompts the user for a configuration PIN if a device is found. The default configuration PIN for all Kunak devices is 1234, but we advise the user to change it during the first configuration. If the device is not found, restart the device, ensure that the USB cable is correctly connected and the driver installed. Use the Windows Device Manager to see if the device is listed among the detected devices (COM port) and then press the Rescan button to see if the device is shown in the drop down menu.

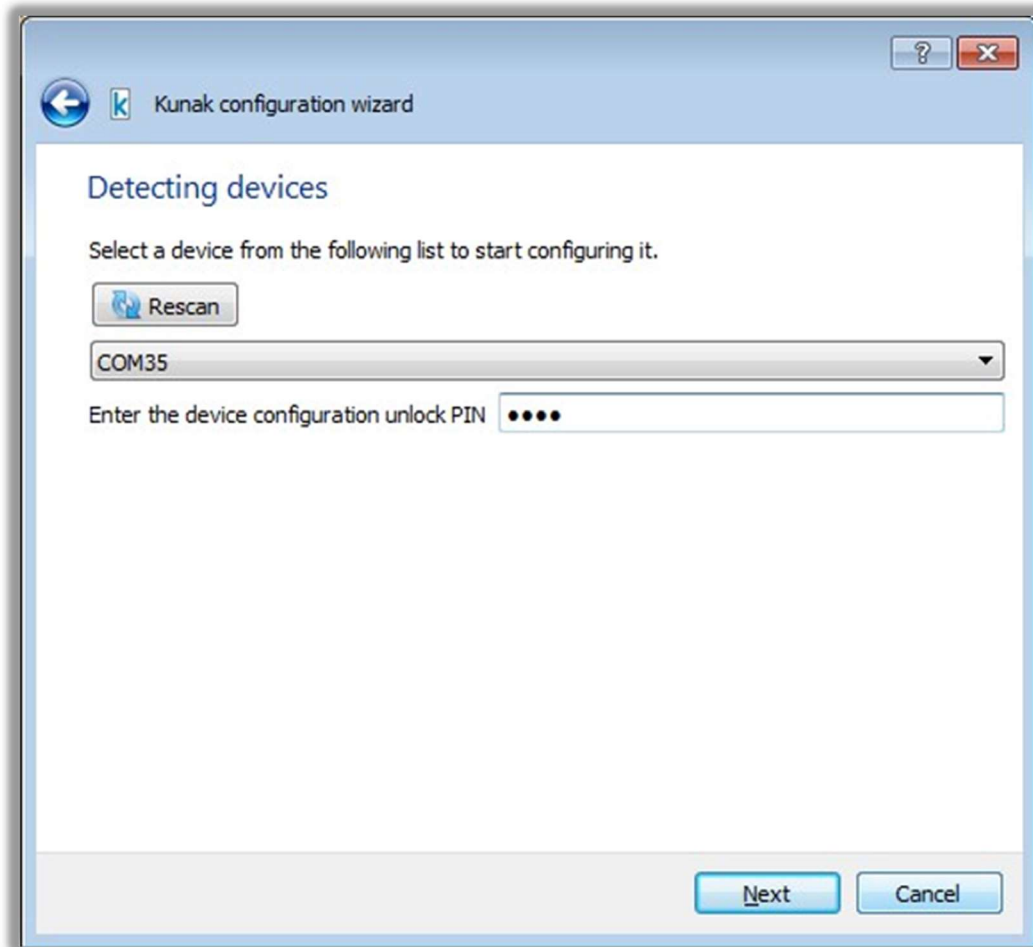


Figure 2 – Device selection

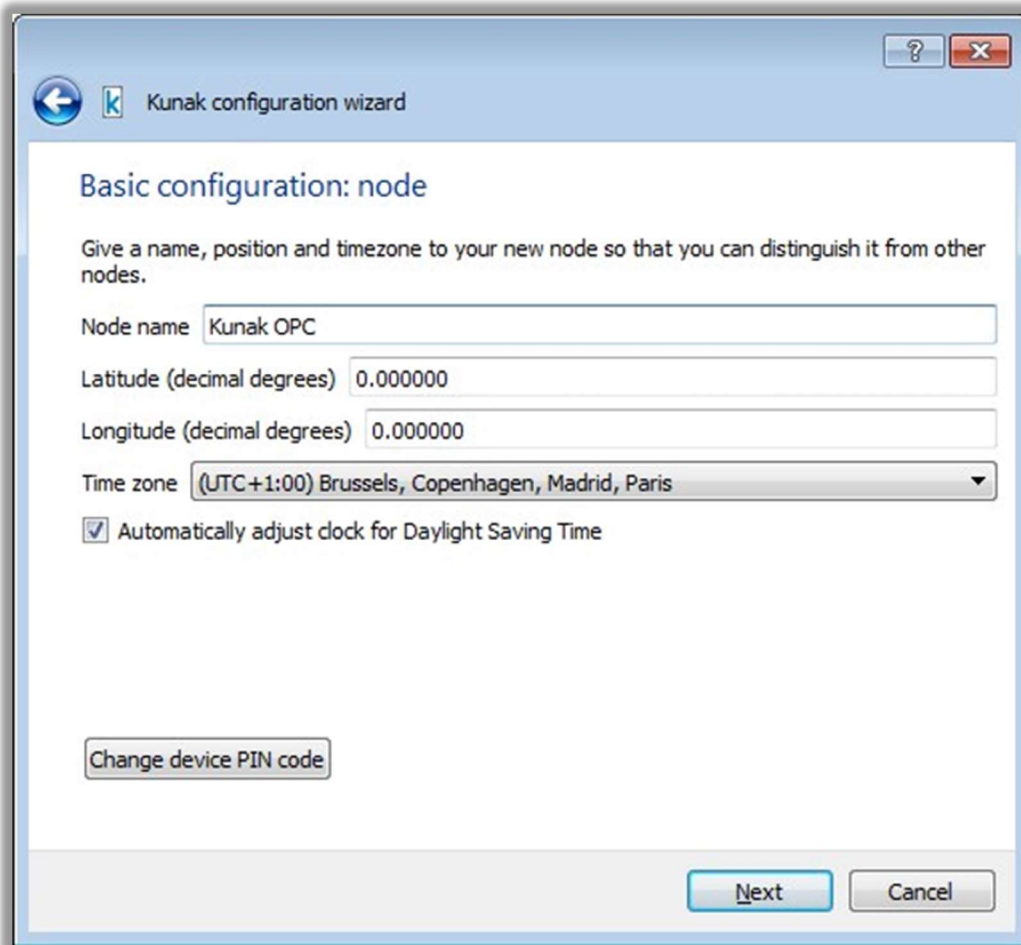
Once the device has been found, press the Next button to setup the device basic configuration.

The 'Basic configuration: node' wizard page, shown below, is a basic configuration screen. In this page, the user has to input the device name. The device name univocally identifies the device and all the samples that will be sent and stored later.

For devices with no GPS receiver, there is the possibility of using an approximate positioning system if the device is equipped with a mobile GPRS data modem. In this case, the position is calculated from data obtained from the GPRS data modem. However, for static devices, it is possible to disable this approximate positioning system by just entering valid values greater than 0.0 for latitude and longitude. Those coordinates are used to display the position of the device in the map section of the Kunak Cloud platform.

The user can also choose the time zone where the device is going to be installed and whether or not the device has to automatically change to Daylight Saving Time. On the lower left corner there is a button to change the current

configuration PIN code to a different one. The PIN code has to have a minimum of 4 characters and a maximum of 8.



Kunak configuration wizard

Basic configuration: node

Give a name, position and timezone to your new node so that you can distinguish it from other nodes.

Node name

Latitude (decimal degrees)

Longitude (decimal degrees)

Time zone

☒ Automatically adjust clock for Daylight Saving Time

Figure 3 – Basic node configuration

The next configurator wizard page ('Basic configuration: connection') is where communications are configured. The proper tab for the device type under configuration will be selected automatically.

2.3.3 Basic configuration: connection

2.3.3.1 K-1xx devices (GPRS)

For the K-1xx devices that have GPRS connectivity, the following parameters have to be configured: the SIM card PIN code, APN, user and password. Those values are usually given by the provider of the SIM card and can also be found online, at least for the main telecommunication operators.

The next two fields: GWWP server URL or IP and GWWP PORT come with preloaded default values.

After that, the user should configure the connection period of the device and the operation mode. Selecting the right operation mode for each application is crucial: a short period of time (lower than 15 minutes) consumes way more power than a longer period of time. The drawback of a longer communication period is that the data is refreshed fewer times than with a shorter period. When an alarm is triggered in any of the connected sensors, the device sends the data regardless the configured period, so rest assured that the alarm will arrive as soon as some seconds after it is triggered.

The working mode radio buttons can be used to select between the two available working modes. When the device is set to operate in normal working mode, it sends all the stored sensor samples in every communication period whereas when set to operate in alarms working mode only an average per sensor

is sent every transmission period. This power saving mode is useful to lower the amount of data to be transmitted over the air and thus lowering the power consumed during the transmission as well.

On the lower left corner there is a check box to avoid the connection test that takes place on the next step. It is recommended to pass the test to ensure that all the data configured in the current page is correct.

Kunak configuration wizard

Basic configuration: connection

Please enter the available communication interfaces different configurations. Beware that high frequency connections drain more battery.

GRPS LoRa WAN WiFi

SIM PIN

SIM APN

APN user

APN password

KWP server URL or IP

KWP port

Connection period

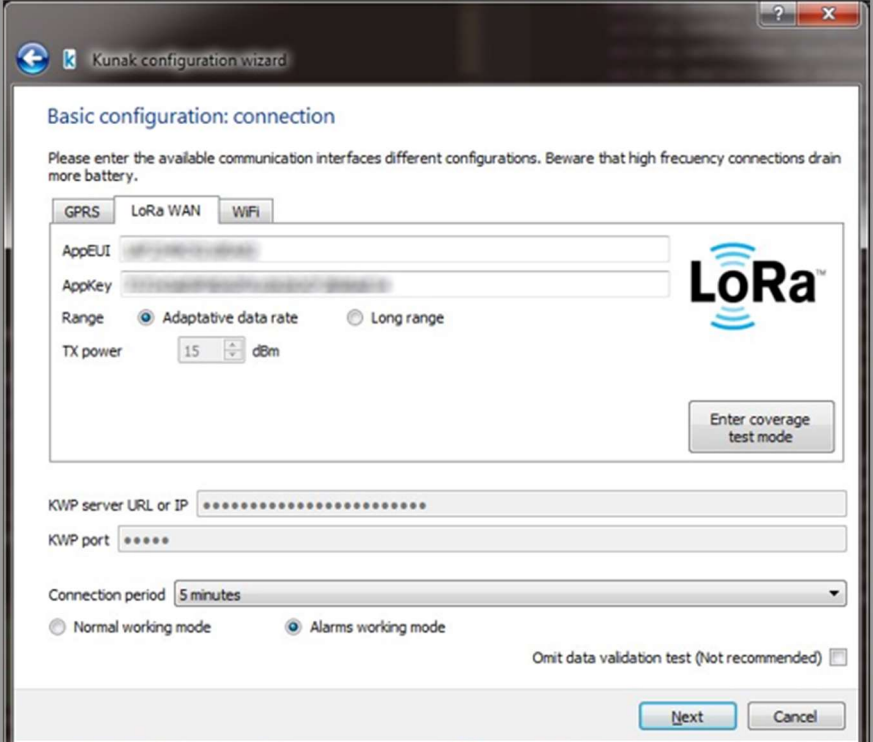
☒ Normal working mode ☐ Alarms working mode

Omit data validation test (Not recommended) ☐

Figure 4 – Basic communications configuration

2.3.3.2 K-2xx devices (LoRa WAN)

K-2xx devices have a LoRa WAN compatible connectivity. For those devices, the gateway AppEUI and AppKey have to be entered. By default, the adaptative data rate is selected and it is recommended. For nodes with extremely low coverage, the Long range mode can be chosen. Beware that only a very small number of devices can be configured in long range mode in the same Gateway due to ETSI standard restrictions. In Long range mode, the TX power can be modified (keep in mind that the maximum PIRE allowed is 14 dBm in the 868 MHz band).



The screenshot shows the 'Kunak configuration wizard' window. The title bar says 'Kunak configuration wizard'. The main window has a tabbed interface with 'LoRa WAN' selected. The 'Basic configuration: connection' section contains the following fields and options:

- AppEUI: [text input field]
- AppKey: [text input field]
- Range: ☒ Adaptive data rate, ☐ Long range
- TX power: [15] dBm (with up/down arrows)
- Enter coverage test mode: [button]
- KWP server URL or IP: [text input field]
- KWP port: [text input field]
- Connection period: [5 minutes] (dropdown menu)
- Normal working mode: ☐ Alarms working mode: ☒
- Omit data validation test (Not recommended): ☐
- Next: [button] Cancel: [button]

The LoRa logo is visible on the right side of the configuration area.

Figure 5 – Basic communications configuration (LoRa equipped devices)

2.3.4 Device status and firmware update

If the data validation test wasn't omitted in the previous section, the next page of the Kunak Configurator will show a brief status report as well as the result of the connection test. If the test is successful, the message 'Device successfully connected to the Internet' should appear meaning that the PIN code, APN, user and password are correctly configured.

This step can be used to access the Kunak Firmware Update application with the button labeled as Program new firmware. This application is explained thoroughly in the next section of this manual.

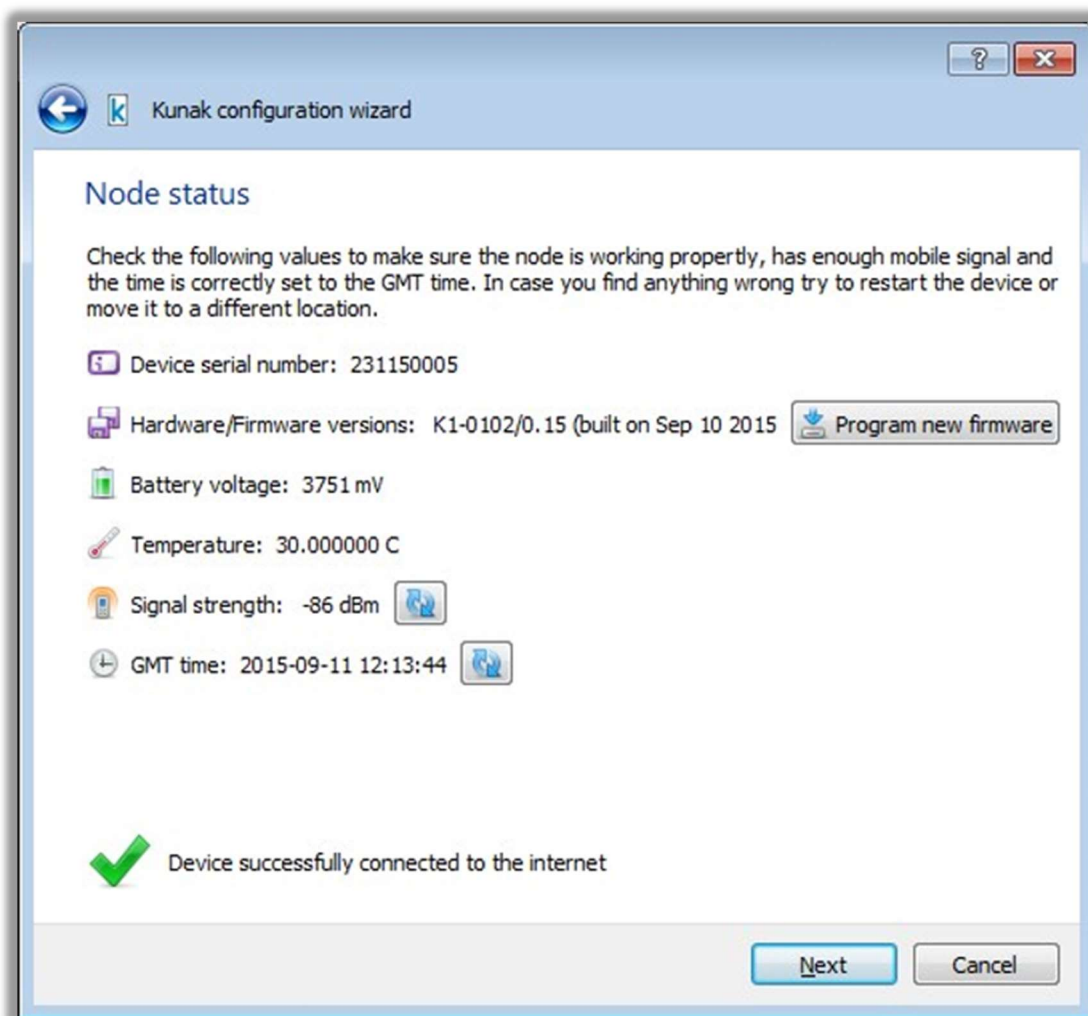


Figure 6 – Node status

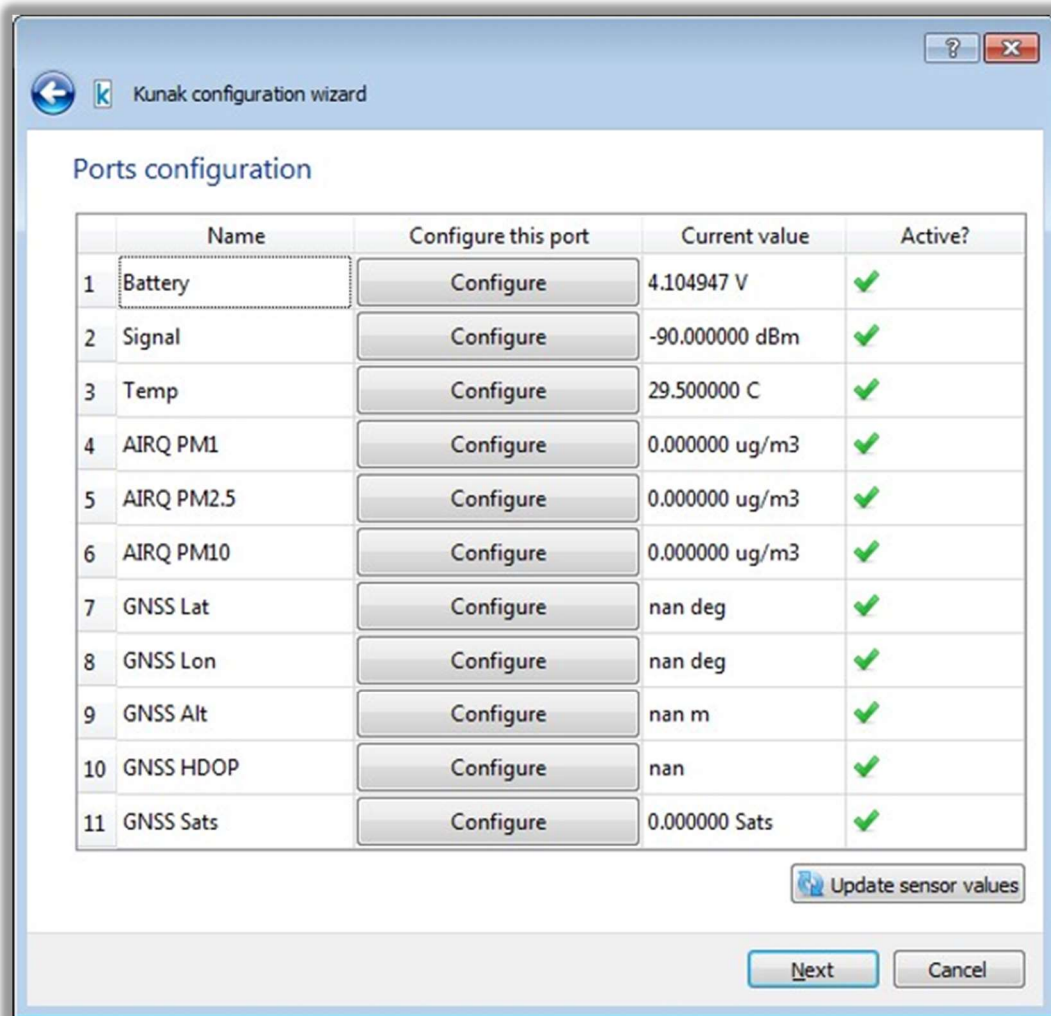
2.3.5 Ports configuration

The following page of the wizard is the ports configuration page. Here, a list of available ports is shown with their names, current values and status. There are four different types of sensors: analog sensors, digital sensors, integrated sensors and expansion sensors.

The digital and analog sensors have to be configured, calibrated and enabled by the user during the first configuration of the device (if they are going to be used).

The integrated sensors are factory calibrated and can't be calibrated through this application. Nevertheless, the user is able to set custom thresholds for these sensors.

The expansion sensors are usually factory calibrated, but the user can freely modify those calibrations as well as sampling periods and alarm thresholds. Although the names are factory set and fixed.



	Name	Configure this port	Current value	Active?
1	Battery	Configure	4.104947 V	✓
2	Signal	Configure	-90.000000 dBm	✓
3	Temp	Configure	29.500000 C	✓
4	AIRQ PM1	Configure	0.000000 ug/m3	✓
5	AIRQ PM2.5	Configure	0.000000 ug/m3	✓
6	AIRQ PM10	Configure	0.000000 ug/m3	✓
7	GNSS Lat	Configure	nan deg	✓
8	GNSS Lon	Configure	nan deg	✓
9	GNSS Alt	Configure	nan m	✓
10	GNSS HDOP	Configure	nan	✓
11	GNSS Sats	Configure	0.000000 Sats	✓

Update sensor values

Next Cancel

Figure 7 – Ports configuration

In order to individually configure each sensor, there is the Configure button on each row of the table. When clicked, a window as the one shown in the following picture appears. Each sensor has different factory default values that can be modified in some cases.

2.3.5.1 General configuration

Usually, the first sensors that appear on the list shown in the figure 6 are analog sensors. To check what kind of sensor is each one, click on the Configure button to show the port configuration dialog.

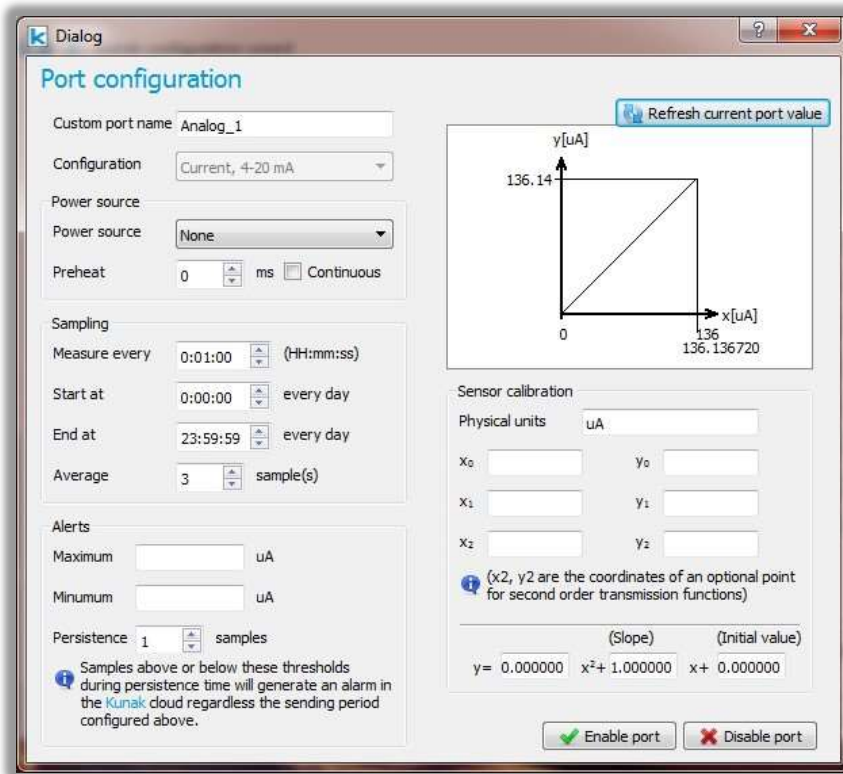


Figure 8 – Port configuration window

Several parameters of a sensor can be configured. Depending on the type of sensor and its configuration some values are editable or not. Below there is a brief explanation on what each value represents and how to configure them:

- Custom port name: It is recommendable to change the sensor name field so that the sensor is easier to find in the future both in the Kunak Cloud web and the Kunak Cloud REST API. For example, if the default sensor name is Digital_1 and the user is going to connect a water meter, the user can name it that way: 'water meter'.
- Configuration: input type supported by the port. Most sensors only have one configuration while others have multiple configurations. Analog sensors are configured to work as voltage or current sensors via hardware and they cannot be switched using this drop down menu, therefore special attention must be taken when connecting analog sensors to the right analog input.
- Power source:
- Power source: configurable if available on the hardware. The K-1xx can have two separate power sources for external sensors; they are labeled VS1 and VS2. Usually VS1 is a 6 Volts power source and VS2 is a 16 Volts power source but that output voltage can be modified upon request when ordering the K-1xx.
- Preheat: a timer that allows the user to configure how much in advance the power source should be turned on before sampling this sensor. Setting the Continuous checkbox forces the selected voltage source to be switched on at all times. Beware that the longer the voltage source is kept on

the more power is drained from the battery and the shorter the lifespan of the battery. If the preheat is set to 0 seconds, the power sources will be powered on some milliseconds before measuring the sensors which can be enough in some cases.

- Sampling
- Measure every: the period between samples of this sensor. The shorter the sampling period, the more times the device has to wake up from deep sleep and the more battery it consumes.
- Start at & end at: the sensor only operates during that intra-day working window thus reducing power consumption the rest of the time. That feature is especially useful when working with solar panels.
- Average: it allows the user to take several samples for each output sample therefore lowering the noise but increasing the needed time to obtain a sample.
- Alerts:
- Max & min: configurable values that make the device instantly inform if the sensor sample is beyond the Maximum or Minimum thresholds
- Persistence: used to avoid false alarms by requiring more than one consecutive sample beyond the thresholds in order to trigger the alarm.
- Sensor calibration: the final and most important step in order to have a sensor working is to calibrate the sensor. This process consists on setting a correspondence between the engineering units of the sensor (uA in the example shown) with the real world units (Celsius degrees in the example shown), which depend on the connected sensor. There are two different options to calibrate a sensor. They are explained below
- Physical units: real world unit/magnitude of this sensor.
- Sensor calibration:
- Via two or three points ($\{x_0, y_0\}$, $\{x_1, y_1\}$, $\{x_2, y_2\}$): Usually, the sensor manual comes with a correspondence table which indicates the correspondence between the input measured by the sensor and the real world unit/magnitude. For example, -20°C is equivalent to 4 mA, -20°C is equivalent to 0 V, 1 pulse is equivalent to 0.3 liters of water, and so on. In that case, the user should enter those values as shown in the following picture:

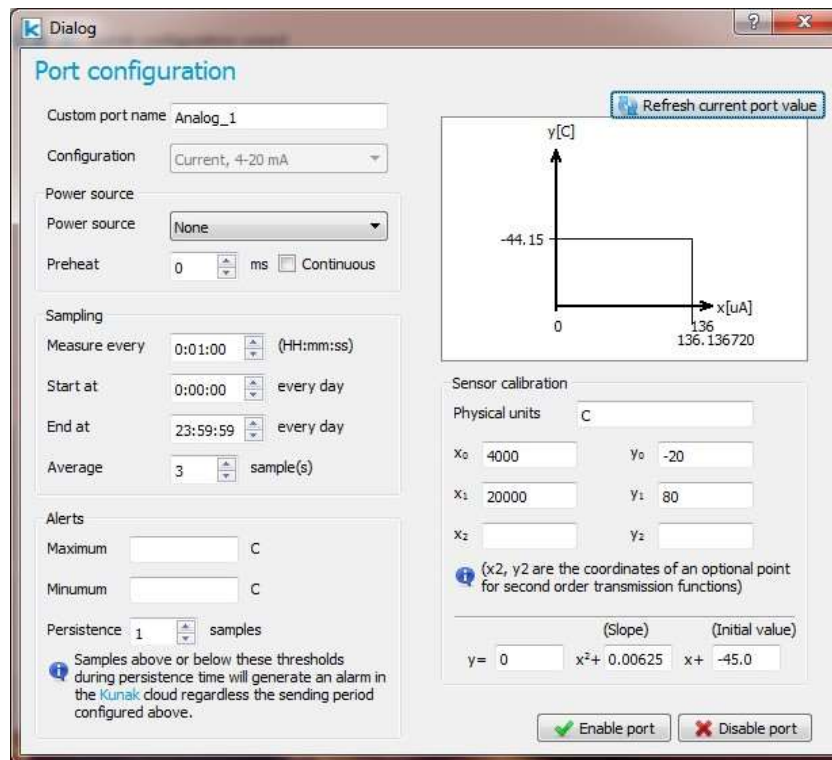


Figure 9 – Port calibration via two or three points

If no table is given by the manufacturer of the sensor or the sensor needs recalibration due to degradation over time; there is also the possibility to enter a known physical value (i.e. 20 °C) to the sensor and write down the output current of the sensor. Then at least a second known physical value is needed in order to finish the calibration (i.e. 25° C) and its corresponding output current. Entering those values automatically generates the conversion polynomial which leads us to the second way of calibrating a sensor: entering the calibration polynomial if given by the manufacturer.

- Entering the calibration polynomial if given by the manufacturer: set the coefficients of the input boxes next to x^2 and x labels and the one below the initial value label.

If the sensor allows user calibrations, the fields are filled with white background while. If the sensor does not allow calibration, those fields appear greyed out.

To calibrate a sensor, the user can refresh ('Refresh current port value' button) the current value and associate a real world value Y to the engineering world value X . The Y value can be obtained using a previously calibrated sensor or a known environment, for example, calibrated gas. In the example of a water meter, the user manual gives the amount of liters per pulse which is the value of the Slope coefficient. In the example of a thermometer that goes from 0 to 10 volts, if the manual says that a value of 0 corresponds to a value of -50 degrees Celsius and a value of 10 volts corresponds to a value of 80 degrees Celsius, the user should use the points calibration as follows: x_0 is 0 V and y_0 is -50 and x_1 is 10 volts and y_1 is 80 Celsius.

Once the calibration process is finished, the last thing to do is to click on the Enable sensor button which closes the window and returns to the list of all ports of the device. There, the current value read by the recently configured sensor should be visible.

2.3.5.2 Configuration of analog 4-20 mA sensors

The following image shows an example of a 4-20 mA port configuration dialog. The analog sensors are configured to work as voltage or current sensors via hardware and they cannot be switched using the Configuration drop down menu, therefore special attention must be taken when connecting analog sensors to the right analog input. In this example, sensor Analog_1 is configured via hardware as a current, 4-20 mA sensor, thus the only possible measure is current. It is also possible to connect 0-20 mA sensors perfectly well.

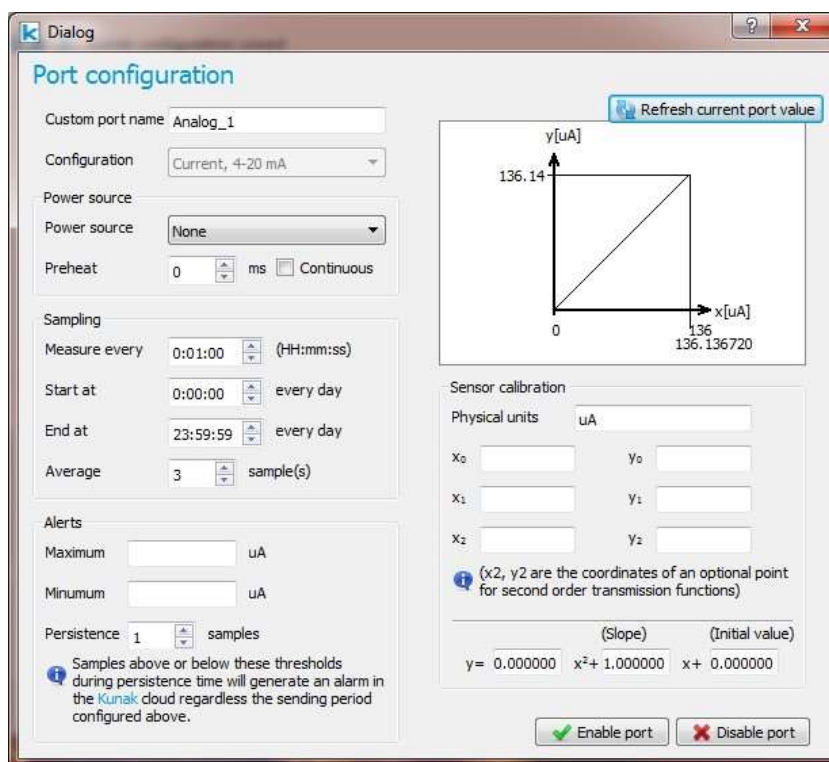


Figure 10 – Port calibration via two or three points

Usually, the sensor manual comes with a correspondence table such as -20°C is equivalent to 4 mA and 80°C is equivalent to 20 mA, in that case, the user should enter those values as shown in the following picture.

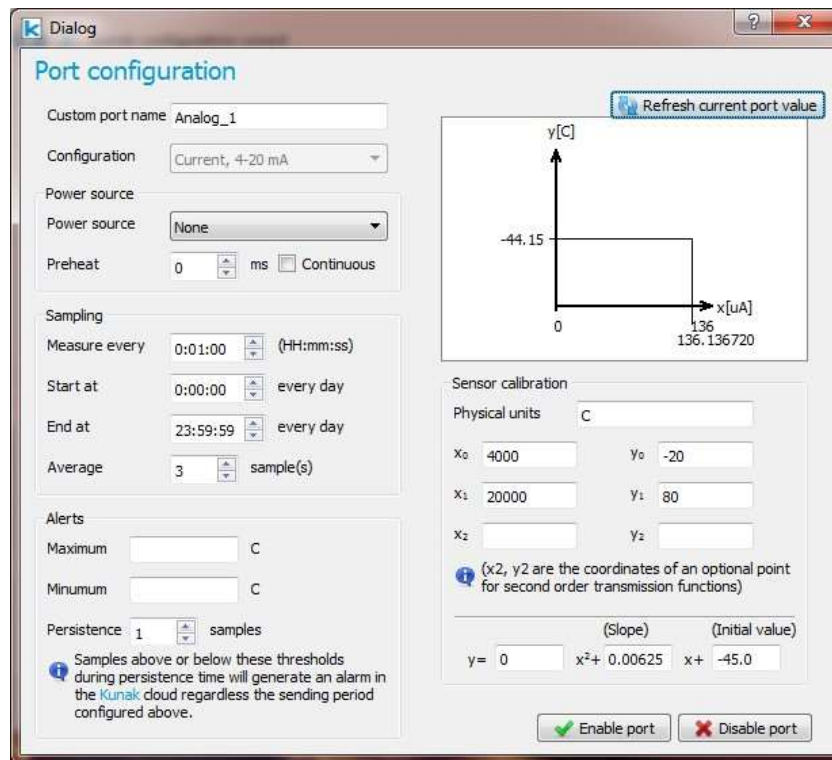


Figure 11 – Calibration of a 4-20 mA sensor

If no table is given by the manufacturer of the sensor or the sensor needs recalibration due to degradation over time; there is also the possibility to enter a known physical value (i.e. 20 °C) to the sensor and write down the output current of the sensor. Then at least a second known physical value is needed in order to finish the calibration (i.e. 25° C) and its corresponding output current. Entering those values automatically generates the conversion polynomial.

If the conversion polynomial is known, it is possible to set directly the coefficients of the input boxes next to x^2 and x labels and the one below the initial value label.



To ensure proper actuator operation the right power source/preheat configuration must be applied.

2.3.5.3 Configuration of analog 0-10 V sensors

The following image shows an example of a 0-10 V port configuration dialog. The analog sensors are configured to work as voltage or current sensors via hardware and they cannot be switched using the Configuration drop down menu, therefore special attention must be taken when connecting analog sensors to the right analog input. In this example, sensor Analog_1 is configured via hardware as a Voltage, 0-10 V sensor, thus the only possible measure is voltage. It is also possible to connect 0-1 V sensors perfectly well due to the high resolution of the ADC.

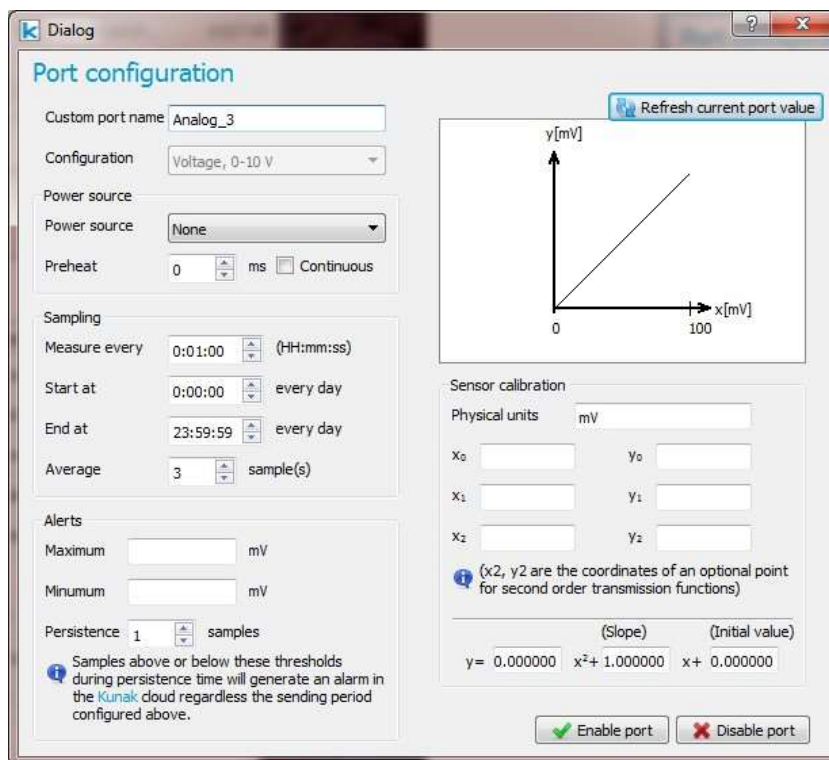


Figure 12 – Analog 0-10 V port configuration

Usually, the sensor manual comes with a correspondence table such as -20°C is equivalent to 0 V and 80°C is equivalent to 10 V, in that case, the user should enter those values as shown in the following picture.

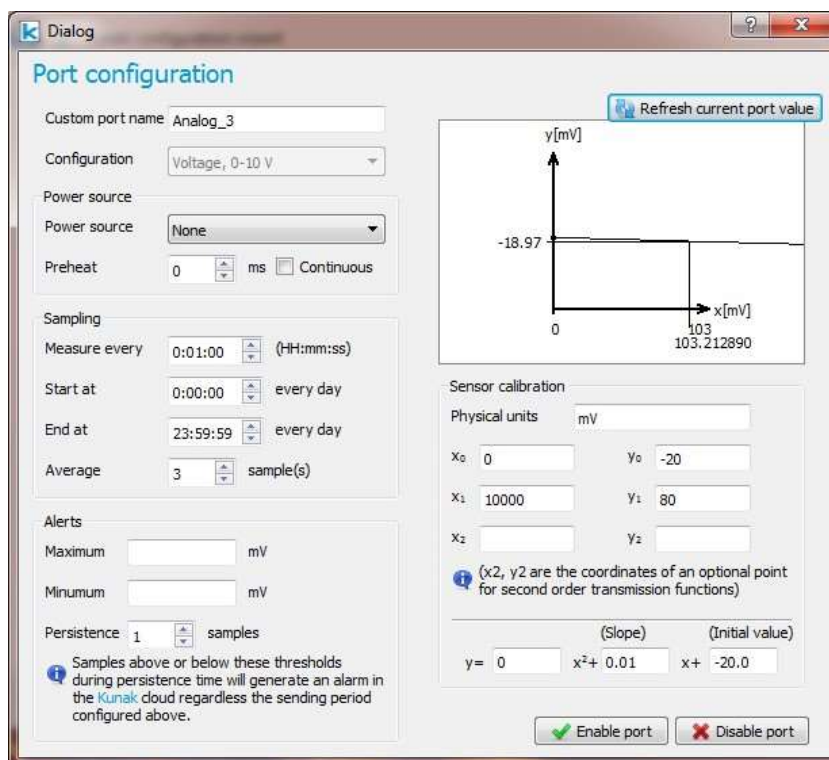


Figure 13 – Calibration of a 0-10 V sensor

If no table is given by the manufacturer of the sensor or the sensor need recalibration due to degradation over time; there is also the possibility to enter a known physical value (i.e. 20 °C) to the sensor and write

down the voltage output of the sensor. Then at least a second known physical value is needed in order to finish the calibration (i.e. 25° C) and its corresponding output voltage.

If the conversion polynomial is known, it is possible to set directly the coefficients of the input boxes next to x^2 and x labels and the one below the initial value label.



To ensure proper actuator operation the right power source/preheat configuration must be applied.

2.3.5.4 Configuration of pulse counter digital sensors

In the case of digital sensors though, it is possible to configure the type of sensor with the Configuration drop down menu. The following figure shows an example configuration of a digital pulse counter (totalizer).

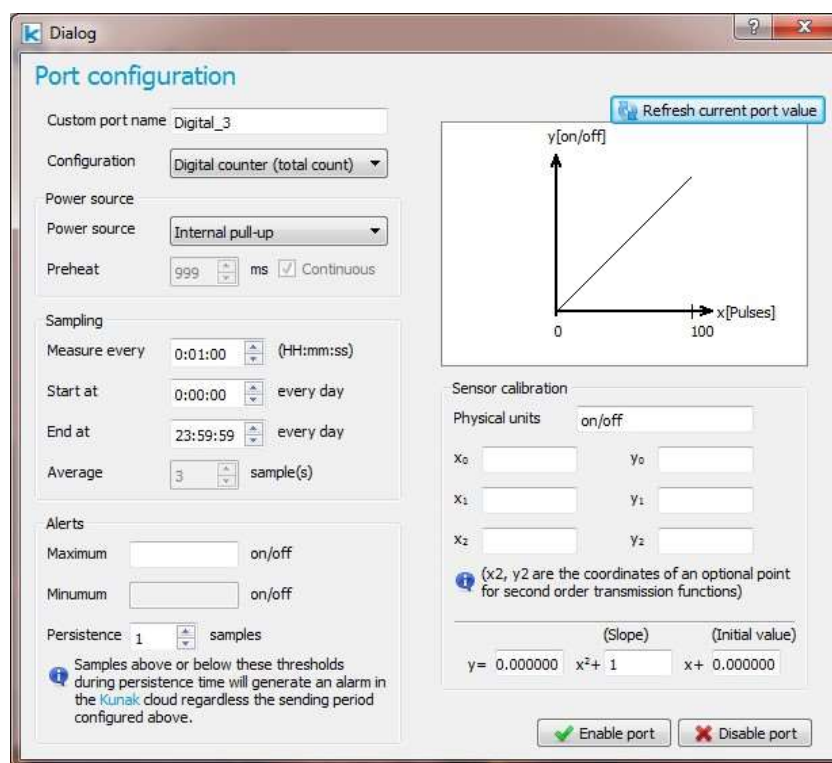


Figure 14 – Digital counter totalizer port configuration

There are two types of pulse counters, the totalizer pulse counters and the partial pulse counters. Totalizer pulse counters maintain a count of the number of pulses since the first configuration whereas the partial pulse counter resets the count every time a sample is taken from the sensor.

In the case of pulse counters it is also very important to calibrate the sensor. Usually, sensor manufacturers specify a relation between one pulse and number of measured units. A typical example would be 1 pulse is equivalent to 0.3 liters of water. These types of calibrations should be entered directly in the calibration polynomial as shown in the picture below.

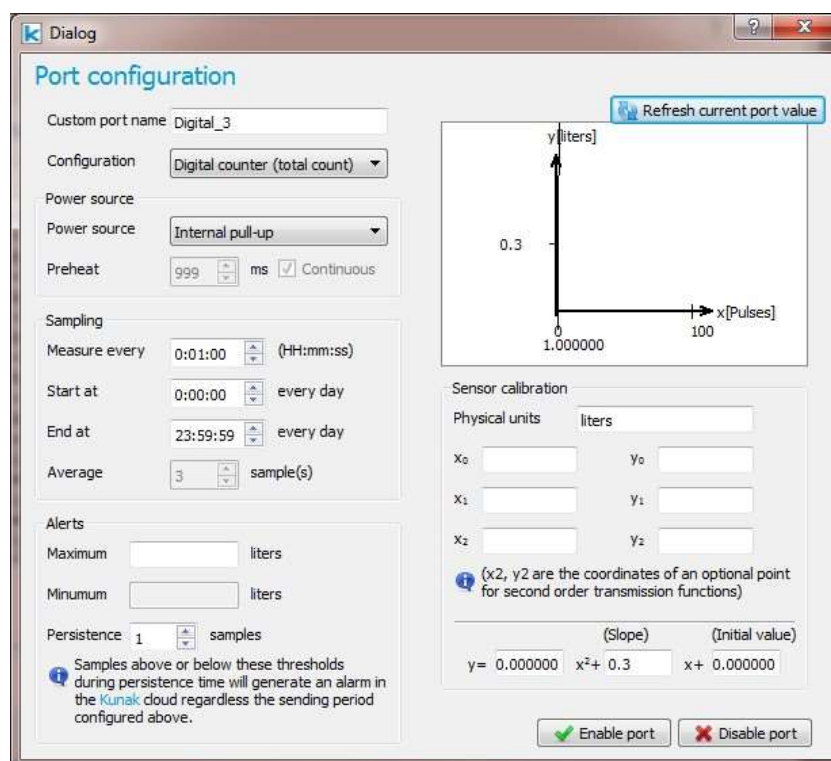


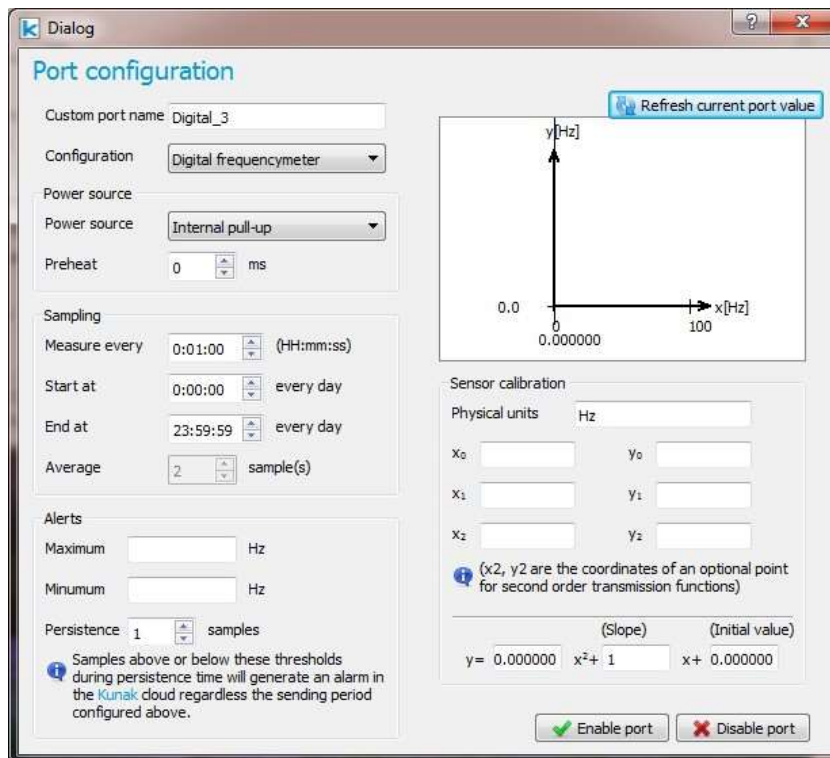
Figure 15 – Calibration of a pulse counter sensor

2.3.5.5 Configuration of frequency meter digital sensors

The following figure shows an example configuration of a digital frequency meter sensor. There are two different frequency meter configurations available: Digital average frequency and Digital instantaneous frequency.

The digital average frequency sensor counts the number of pulses between two samples and divides that number of pulses by the time between the two samples. Therefore, low frequencies can be measured accurately with several Hz decimal precision.

The digital instantaneous frequency sensor configuration, on the other hand, allows to count the instantaneous frequency. That frequency is internally calculated and updated each second, providing high frequency samples up to 3.2 kHz.



Port configuration

Custom port name: Digital_3

Configuration: Digital frequencymeter

Power source: Internal pull-up

Preheat: 0 ms

Sampling:

Measure every: 0:01:00 (HH:mm:ss)

Start at: 0:00:00 every day

End at: 23:59:59 every day

Average: 2 sample(s)

Alerts:

Maximum: Hz

Minimum: Hz

Persistence: 1 samples

Samples above or below these thresholds during persistence time will generate an alarm in the Kunak cloud regardless the sending period configured above.

Sensor calibration:

Physical units: Hz

x_0 : y_0

x_1 : y_1

x_2 : y_2

(x_2 , y_2 are the coordinates of an optional point for second order transmission functions)

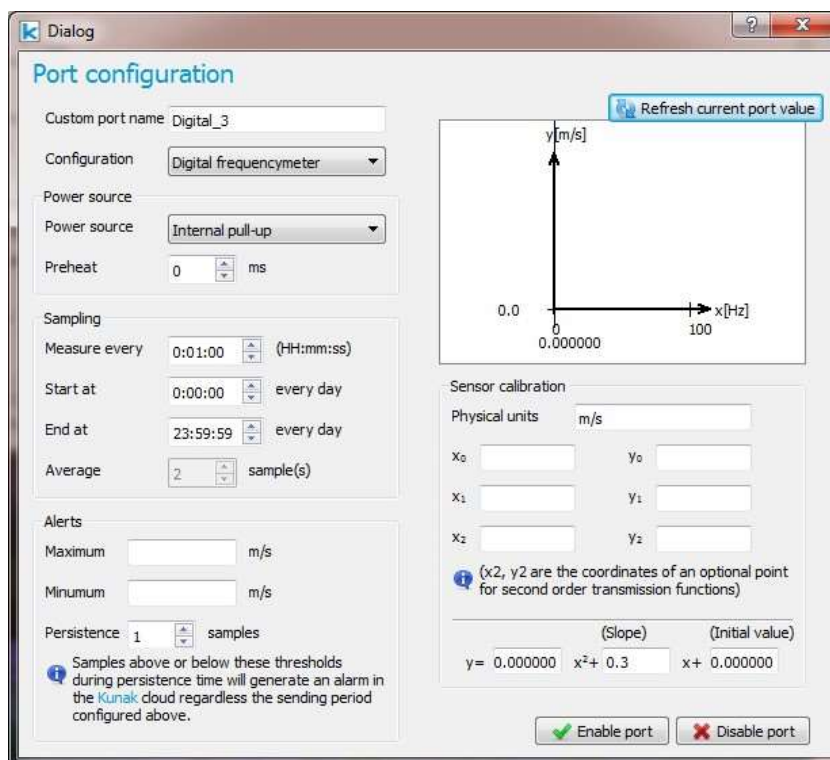
(Slope) (Initial value)

$y = 0.000000 x^2 + 1 x + 0.000000$

Enable port Disable port

Figure 16 – Digital counter totalizer port configuration

Usually, sensor manufacturers specify a relation between the frequency and the number of measured units. A typical example would be 1 Hz is equivalent to 0.3 m/s of wind speed. This type of calibrations should be entered directly in the calibration polynomial as shown in the picture below.



Port configuration

Custom port name: Digital_3

Configuration: Digital frequencymeter

Power source: Internal pull-up

Preheat: 0 ms

Sampling:

Measure every: 0:01:00 (HH:mm:ss)

Start at: 0:00:00 every day

End at: 23:59:59 every day

Average: 2 sample(s)

Alerts:

Maximum: m/s

Minimum: m/s

Persistence: 1 samples

Samples above or below these thresholds during persistence time will generate an alarm in the Kunak cloud regardless the sending period configured above.

Sensor calibration:

Physical units: m/s

x_0 : y_0

x_1 : y_1

x_2 : y_2

(x_2 , y_2 are the coordinates of an optional point for second order transmission functions)

(Slope) (Initial value)

$y = 0.000000 x^2 + 0.3 x + 0.000000$

Enable port Disable port

Figure 17 – Calibration of a frequency sensor

2.3.5.6 Configuration of opened/closed circuit sensors

The last type of digital sensor configuration is the opened / closed circuit which is shown in the following figure.

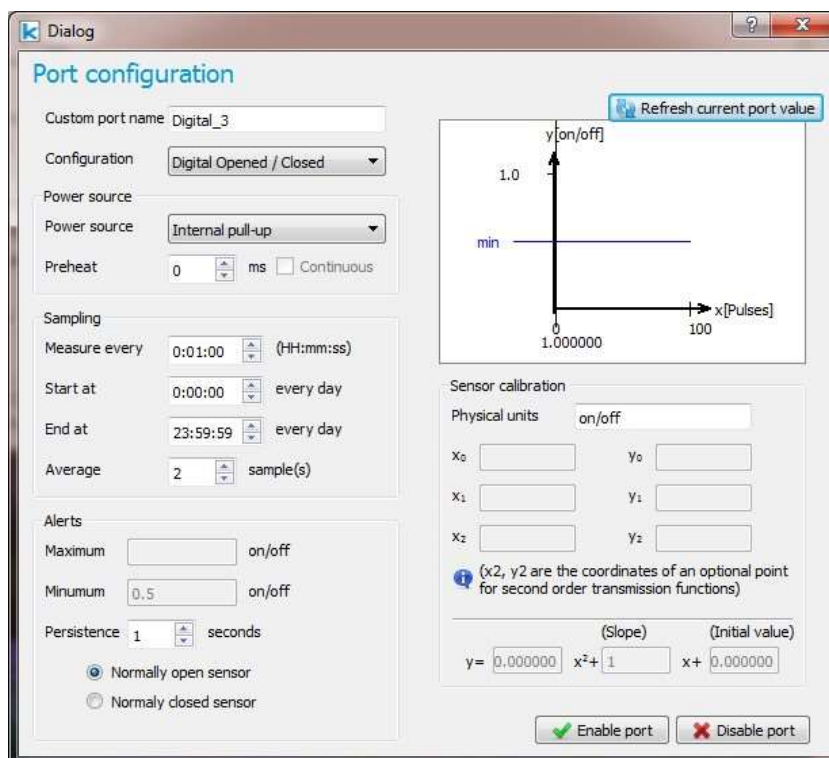


Figure 18 – Opened/closed port configuration

This type of sensors does not have physical units; they can have the value 1 or the value 0. Usually, the sensor has a stable status and a status that should generate an alarm. The available alarm options for this type of configuration are normally closed sensor and normally opened sensor. The first option would generate an alarm as soon as the circuit is opened and the second option would generate an alarm as soon as the circuit is closed.

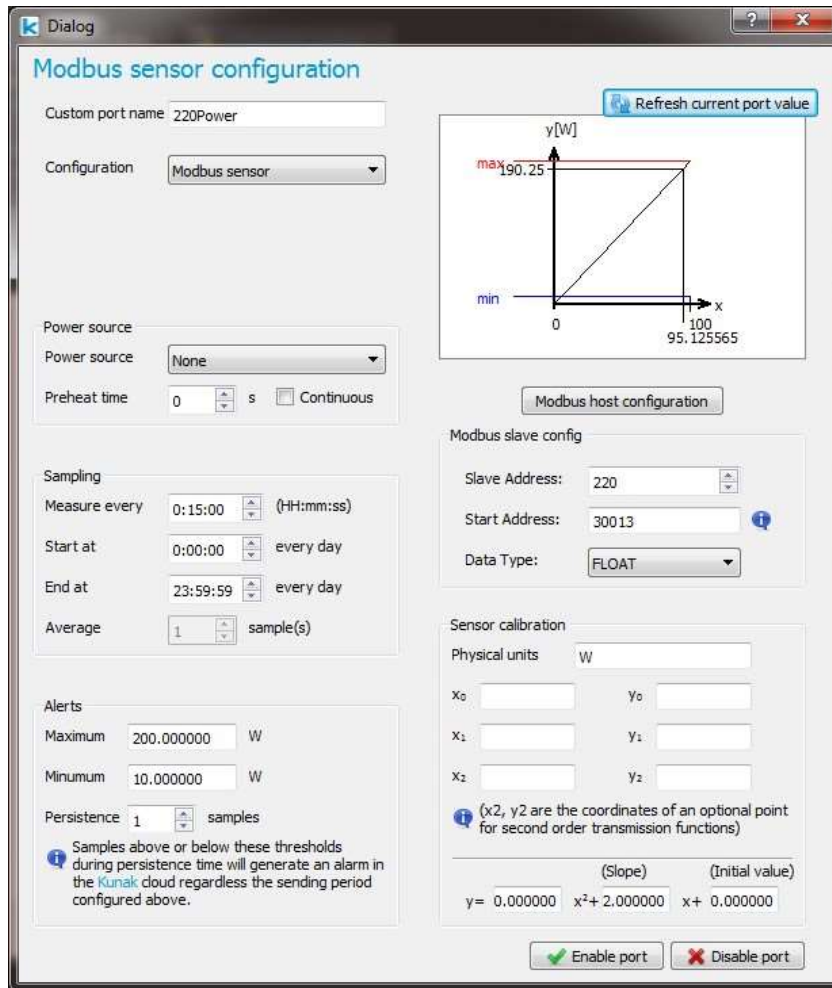
No calibration is usually necessary for this configuration.

2.3.5.7 Configuration of integrated or expansion sensors

The integrated and expansion sensors are factory configured and they do not need any intervention from the user.

2.3.5.8 Configuration of Modbus sensors

Modbus ports can query any connected Modbus compatible slave for a registry value as it is done with other sensors of the device. Thus, power source, sampling period and working window, alerts and calibrations apply as in previous cases. The following image shows the advanced Modbus sensors configuration dialog.



The dialog box is titled "Modbus sensor configuration". It contains several sections for configuring a Modbus sensor:

- Custom port name:** 220Power
- Configuration:** Modbus sensor (dropdown)
- Power source:** None (dropdown)
- Preheat time:** 0 s, with a checkbox for Continuous.
- Sampling:**
 - Measure every: 0:15:00 (HH:mm:ss)
 - Start at: 0:00:00 every day
 - End at: 23:59:59 every day
 - Average: 1 sample(s)
- Alerts:**
 - Maximum: 200.000000 W
 - Minimum: 10.000000 W
 - Persistence: 1 samples
 - Info: Samples above or below these thresholds during persistence time will generate an alarm in the Kunak cloud regardless the sending period configured above.
- Modbus host configuration:**
 - Slave Address: 220
 - Start Address: 30013
 - Data Type: FLOAT
- Sensor calibration:**
 - Physical units: W
 - Points: X0, Y0, X1, Y1, X2, Y2
 - Info: (x2, y2 are the coordinates of an optional point for second order transmission functions)
 - Equation: $y = 0.000000 x^2 + 2.000000 x + 0.000000$
 - Buttons: Enable port, Disable port
- Graph:** A plot of y[W] vs x. The y-axis has a "max" value of 190.25 and a "min" value. The x-axis has a value of 100 and a label 95.125565. A diagonal line represents the calibration curve.

Figure 19 – Advanced Modbus configuration dialog

This dialog allows to configure all the aforementioned generic sensor parameters as well as the specific parameters for Modbus sensors, which are slave address, register start address and register data type.

Slave address is an integer value ranging from 1 to 247 that univocally identifies the slave in the Modbus bus. This allows the user to connect multiple slaves to a single shared bus, therefore, multiple registers from multiple slave devices can be read with only one Kunak device.

Register start address identifies the memory register among the slave registers that the user wants to associate with the current sensor. There are 4 possible read commands available:

- Read coil register (Boolean): Coil start addresses range from 1 to 9999.
- Read discrete input register (Boolean): Discrete input start addresses range from 10001 to 19999.
- Read input register: Input registers start addresses range from 30001 to 39999.
- Read holding register: Holding registers start addresses range from 40001 to 49999.

In Modbus protocol specification, slave's memory registers are mapped inside the frame to a function code and address tuple. This representation is frequently used in devices documentation, and may lead to improper configuration of Modbus registers, therefore in the following section the equivalence between Kunak Modbus Register Configuration and Modbus frames PDU.

Modbus registers are defined on a 1-offset base with respect to Modbus addresses. This means that register 0001 corresponds to address 0000.

In Modbus terminology the representation of registers with base 0 is usually referred as 'Protocol Mode', while the one with base one is commonly referred as 'PLC Mode'.

Modbus registers leftmost number specifies the type of register and it is mapped to Modbus function codes as follows:

- Read coil register (range 0xxxx) function code: 1
- Read discrete input register (range 1xxxx) function code: 2
- Read input register (range 3xxxx) function code: 4
- Read holding register (range 4xxxx) function code: 3

To resume, if the slave device documentation describes Modbus protocol in term of registers we will just type the register we are interested in as is in the configurator dialog box, instead if the Modbus protocol is described in terms of addresses and function codes we will have to apply the aforementioned rules to convert them into the proper register number to type in the configurator dialog box.

Example: read address 0000 function code 3 becomes, register 40001 in the configurator.

Data type configuration informs the Kunak device about how to convert the read word or words into the actual value. For example, when choosing float data type, the K-1xx automatically fetches two 16 bits registers starting from the start address on and then converts that two 16 bits values into a floating point value. That floating point value is the sample that is time stamped, stored and sent as the sensor value.

Additionally, all Modbus ports of the Kunak device share what is called Modbus host configuration. That configuration determines the data rate and parity at which the bus is working. All slave sensors connected to a Modbus bus must be configured at that same data rate and parity, otherwise, malfunction will very likely happen. Most Modbus devices allow the user to configure custom data rates and parities even though they come with a factory default value. Some Modbus devices have a built in screen and buttons that allow such configuration whilst other do not and have to be configured via writing specific Modbus registers (Specific slave device manuals should be consulted for further details).

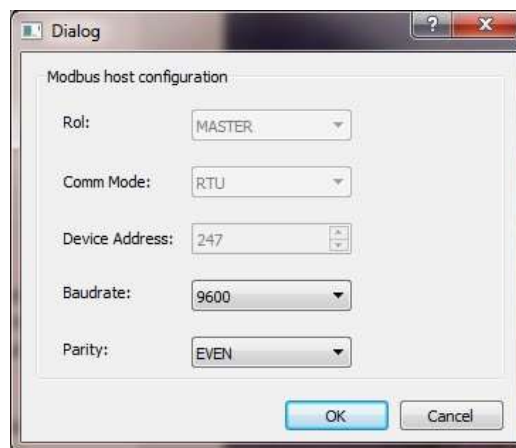


Figure 20 – Modbus host configuration dialog

2.3.5.9 Configuration of actuation ports

Actuation ports allow the user to enable or disable devices remotely. The available configurations for an actuator port are relay and latch.

All actuators can use the embedded power sources if necessary as if they were a regular sensor. This allows the actuator to be powered from the Kunak device.



To ensure proper actuator operation the right power source/preheat configuration must be applied.

When an actuator is configured as a relay, the only additional specific configuration is the default status, that is the state at which the actuator goes when disabled.

When an actuator is configured as a latch, the Latch configuration group box is enabled and lets the user configure the latch pulse duration in ms. That box also shows the second actuator that is used for the second latch pulse. Note that latch configuration will be only available if there is a second actuator available.

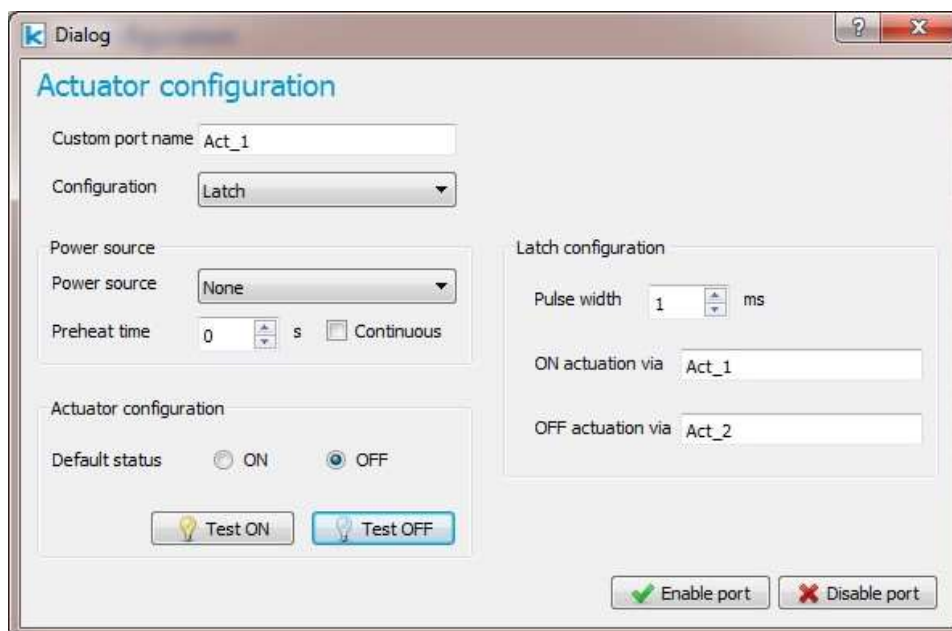


Figure 21 – Advanced actuator configuration

When Act_1 is configured as latch, Act_2 is grayed out meaning that it is being used by Act_1 and thus not configurable for other tasks.

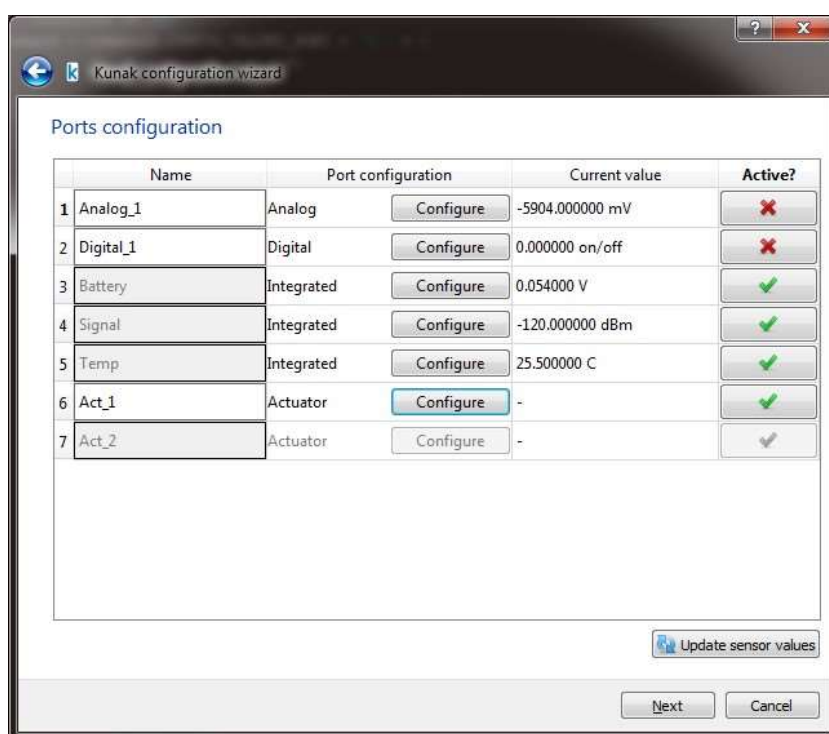


Figure 22 – Example of Act_2 disabled due to Act_1 being configured as latch

Once all the sensors are configured as necessary, press the Next button and the Finish button to proceed with the node allocation process.

2.4 Kunak Firmware Update

The Kunak Firmware Update application is a flashing tool that can be used to flash a new firmware to the device. The application is accessible through the Status screen of the Configurator desktop app or through the start menu icon. In the following figure there is a screenshot of the application.

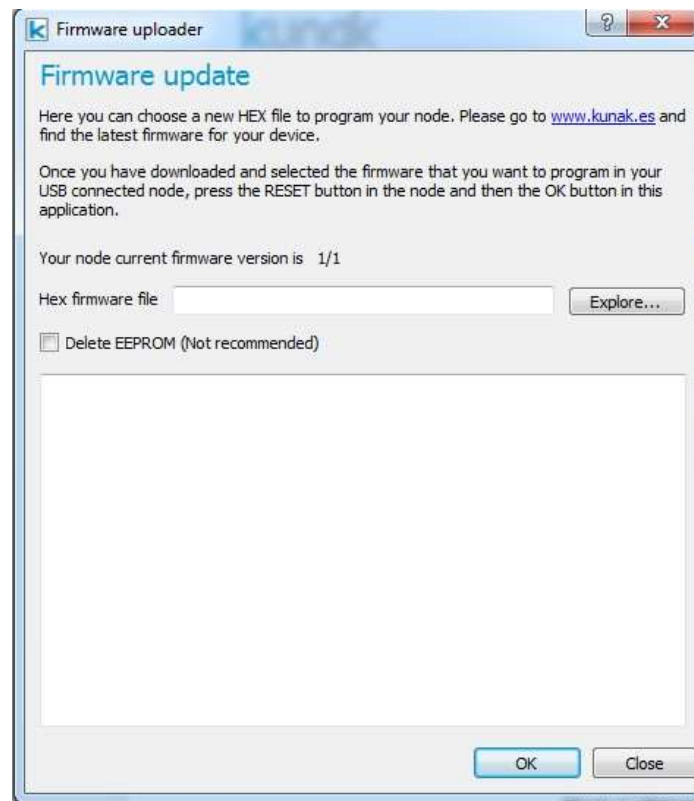


Figure 23 – Firmware update application

Once the application is opened, the user has to follow the steps listed below in order to flash a new firmware:

1. Make sure the device has been charging for a while (1 hour minimum) before performing this procedure.
2. Stop charging the device.
3. Unplug the battery cable from the internal Kunak kernel (check the device manual).
4. Keep the RST button pushed while plugging the battery cable again to enter bootloader mode. None of the LEDs should be bright. If any LED is on, restart the procedure from the beginning and ensure that the device is not being charged during this process. In device manager, there should be a group 'Atmel USB Devices' including a device 'ATxmega256A3BU' if drivers were successfully installed.
5. Once the device is in bootloader mode, plug the USB cable to the Kunak internal kernel and to the computer.
6. Select the Hex firmware file with the button labelled Explore in the application.

7. In case the memory EEPROM has been corrupted, it will be solved by clicking on the Delete EEPROM checkbox.
8. Press the OK button and wait for the process to end. This may take up to one minute. If the procedure ends successfully, the following result must appear on screen:

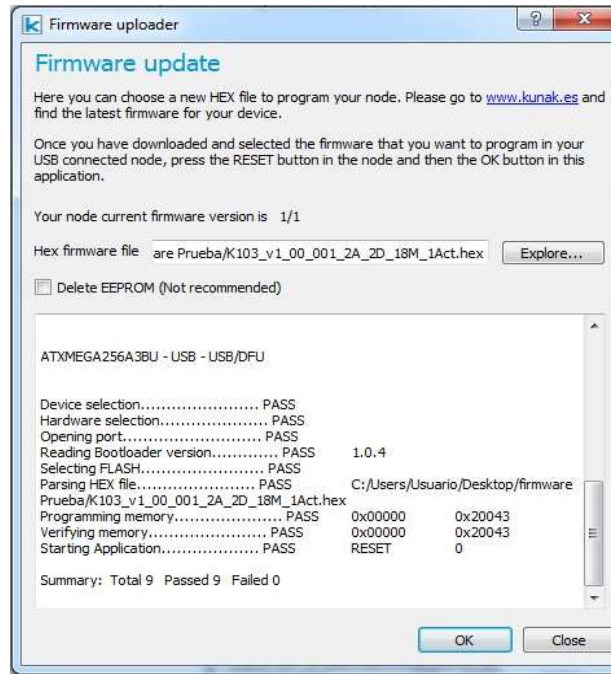


Figure 24 – Firmware successfully loaded

If the device is not in bootloader mode, the USB cable is not plugged or the driver of the bootloader is not installed, the following result is displayed. Check the three possible causes of the error and try again.

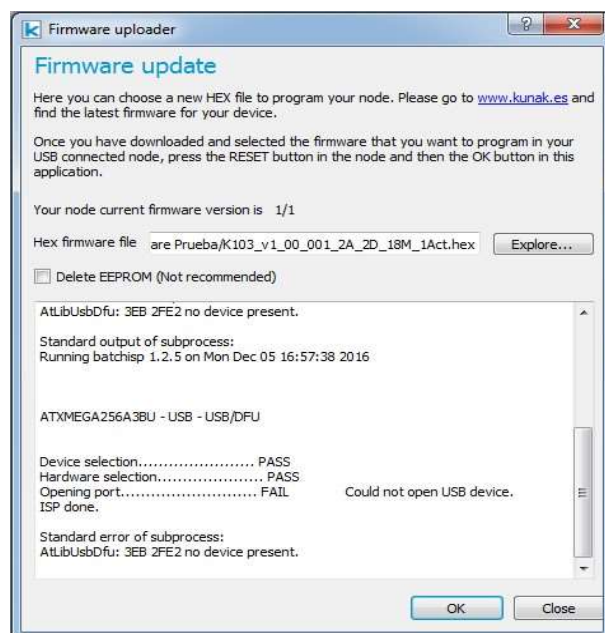


Figure 25 – Firmware not loaded due to device not detected

2.5 KunakCloud

Most of the previously configured settings can be freely modified during run-time via the Kunak Cloud menus. If a configuration is modified online, the new parameters take effect as soon as the configured device connects to the platform (usually in less than a device connection period plus a mobile communications delay).

For detailed information, the 'Kunak Cloud Web' manual can be found on the web section 'Support' (top right), accessible through the following URL: https://kunakcloud.com/user_area/support.jsp.